TEST CODE **000574**

FORM TP 99218

PILOT/MAY 1999

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

MATHEMATICS

UNIT 2 – PAPER 02

$2\frac{1}{2}$ hours

03 JUNE 1999 (a.m.)

This examination paper consists of THREE sections: Module 2.1, Module 2.2, and Module 2.3.

Each section consists of 2 questions.
The maximum mark for each section is 40.
The maximum mark for this examination is 120.
This examination paper consists of 6 pages.

INSTRUCTIONS TO CANDIDATES

1. **DO NOT** open this examination paper until instructed to do so.

2. Answer **ALL** questions from the **THREE** sections.

3. Unless otherwise stated in the question, all numerical answers **MUST** be given exactly **OR** to three significant figures as appropriate.

Examination materials

Mathematical formulae and tables
Electronic calculator
Ruler and compass

SECTION 1 (MODULE 2.1)

Answer ALL The Questions.

1. In a certain chemical reaction, the rate of production of a substance is directly proportional to the amount, x grams, already produced.

(a) (i) Write down a differential equation relating x and the time t, measured in seconds. [1 mark]

(ii) By solving the differential equation in (i), show that $x = Ae^{kt}$, where A and k are constants. [4 marks]

(iii) Given that $k = 0.005$, calculate the time taken for the amount of the substance to triple. [3 marks]

The substance is now removed at a constant rate of h grams per second.

(b) (i) Show that the differential equation for this present condition is given by $dx/dt = kx - h$, where h is a constant. [1 mark]

(ii) Given that at the beginning of the reaction there were p grams of the substance, determine the amount of the substance at time t seconds. [6 marks]

(iii) In one sentence, write what happens to the amount of substance if $kp \leq h$. [1 mark]

(c) Sketch graphs on the same rectangular coordinates system showing the variation with respect to time t, of the amount of substance x, for the cases,

(i) $kp \leq h$ [2 marks]

(ii) $kp > h$. [2 marks]

2. (a) Let sin u = x and cos v = x, where u and v are acute angles.

(i) Prove that $u + v = \frac{\pi}{2}$. [3 marks]

(ii) Show that $\frac{du}{dx} = \frac{1}{\sqrt{1 - x^2}}$. [4 marks]

(iii) Hence, find $\frac{dv}{dx}$. [2 marks]

(b) (i) Sketch, on the same rectangular coordinates system, the exponential and logarithmic functions f and g, respectively, given by

$f(x) = e^x$ and $g(x) = \ln x$.

Label and name the coordinates of points where the graphs intersect the axes. [4 marks]

(ii) State the relationship between f and g. [2 marks]

Given that the derivative at x of the exponential function e^x is e^x,

(iii) find the derivative at x of the logarithmic function ln x [3 marks]

(iv) hence, find the derivative at x of the function f(x)g(x). [2 marks]

SECTION 2 (MODULE 2.2)

Answer ALL The Questions.

3. (a) (i) Prove by mathematical induction that

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

for all positive integers n. **[7 marks]**

(ii) Hence, or otherwise, find in terms of n the sum of the cubes of the first n even positive integers. **[5 marks]**

(b) (i) An arithmetic series has n^{th} term $a + (n - 1)d$, where a and d are real constants. Prove that the n^{th} partial sum, S_n, of the series is given by

$$S_n = \frac{n}{2}\{2a + (n-1)d\}.$$

[3 marks]

(ii) Hence, express

$$\ln (2 \times 2^2 \times 2^3 \times \ldots \times 2^{48} \times 2^{49})$$

in the form k ln 2, where k is an integer. **[5 marks]**

4. (a) With the aid of a suitably labelled diagram, explain the use of the Newton-Raphson method for finding successive approximations of differentiable functions. **[4 marks]**

(b) The function f is given by $f(x) = x^3 + x - 3$.

Show that

(i) f is a strictly increasing function **[3 marks]**

(ii) the equation $f(x) = 0$ has a real root α in the interval $[1,2]$ **[3 marks]**

(iii) the equation $f(x) = 0$ has no other root in the interval $[1,2]$ **[3 marks]**

(iv) by choosing $x_0 = 1$ as an initial approximation to the root α, that a better approximation of α is $x_1 = 1.25$. **[2 marks]**

(c) Let x_n be the n^{th} approximation of α. What is x_{n+1} in terms of x_n? **[5 marks]**

SECTION 3 (MODULE 2.3)

Answer ALL The Questions.

5. (a) The following data show the reactions to a certain drug of a group of persons who participated in a large-scale experiment.

Reactions to Drug

Sex	Severe Reaction	Mild Reaction	No Reaction
Male	520	380	350
Female	330	480	300

A person is randomly selected from the group. What is the probability that the person selected,

(i) has no reaction to the drug? [2 marks]

(ii) is female and has a mild reaction to the drug? [2 marks]

(iii) is male or has a severe reaction to the drug? [4 marks]

(iv) does not have a severe reaction to the drug, given that the person is female? [4 marks]

(b) A bag contains two blue marbles and three red marbles. A marble is drawn at random. If this marble is blue, it is put back in the bag along with another blue marble and a second marble is drawn. However, if the first marble drawn is red, it is removed from the bag and not replaced, and a second marble is drawn.

(i) Draw a tree diagram to represent the selection process described above. [3 marks]

(ii) What is the probability that the second marble drawn is red? [2 marks]

(iii) If the second marble drawn is red, what is the probability that the first marble drawn was also red? [3 marks]

6. (a) A fish pond is constructed for breeding groupers only. The number of groupers in the fish pond at time t (measured in days), can be modelled by

$$F(t) = \frac{500}{(1+49e^{-0.2t})}.$$

(i) How many groupers were in the pond initially? **[2 marks]**

(ii) How quickly is the population of groupers increasing after 5 days? **[3 marks]**

(iii) What is the greatest number of groupers the pond can support? **[3 marks]**

(b) Write down a differential equation to represent each of the following. Carefully define all variables used.

(i) The population of a Caribbean island increases by 3.5% every year. **[2 marks]**

(ii) The number of rabbits on a farm would be twice as many every 40 days except for the fact that 5 rabbits are caught and sold every day. **[3 marks]**

(c) In modelling, a mathematician usually poses a **series of questions** (Q) that need to be answered before model formulation can begin, then lists **factors** (F) that can affect the outcome together with any **assumptions** (A). Finally, the **problem statement** (P) is given in words and in symbols.

Consider the following problem.

A new housing estate has been built and a bus stop has to be placed along the main road that passes through the estate. Where should the bus company place the bus stop?

Write suitable examples of:

(i) 2 questions (Q) **[2 marks]**

(ii) 4 factors (F) **[2 marks]**

(iii) 1 assumption (A) **[1 mark]**

(iv) problem statement (P). **[2 marks]**

END OF TEST